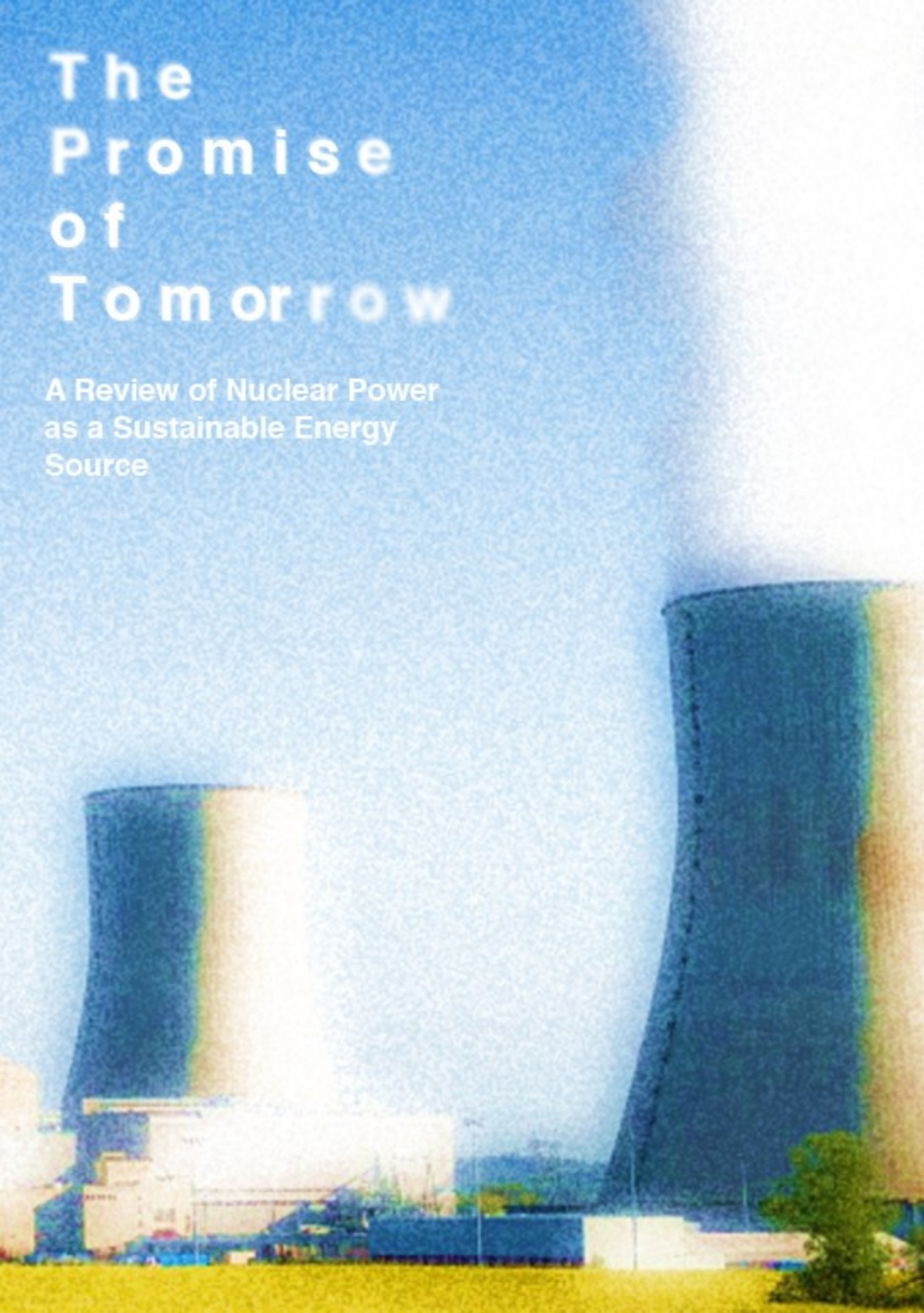


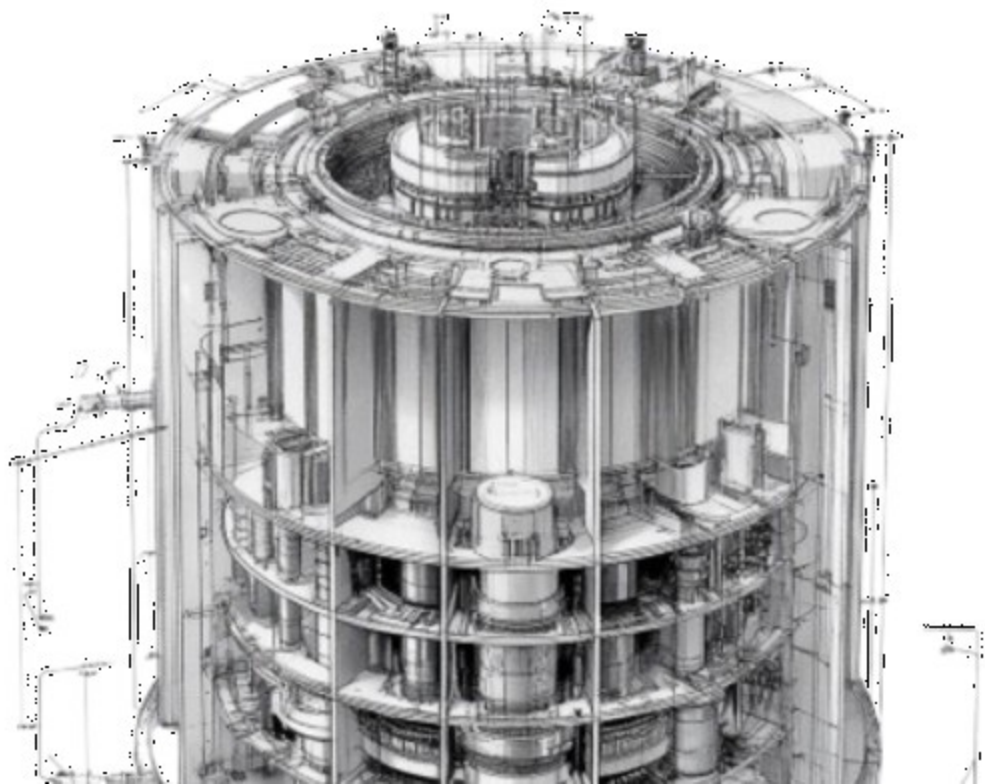
The Promise of Tomorrow

A Review of Nuclear Power
as a Sustainable Energy
Source



A Review of Nuclear Power as a Sustainable Energy Source

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Abstract.

Nuclear energy has long occupied a rather peculiar place in the energy conversation: it is powerful, low-carbon, reliable, and yet often surrounded by questions of safety, waste, and cost. This paper asks a simple question: how viable is nuclear power as a sustainable energy source? To answer it, we first establish how nuclear reactors generate electricity and how major reactor designs differ. We then examine the environmental case for nuclear energy, particularly its low life-cycle carbon emissions, reliability, land efficiency, and the persistent concerns surrounding radioactive waste and nuclear accidents. The discussion then turns to economics, using cost- and value-based measures such as LCOE and LVOE, before finally considering the future of nuclear power through advanced reactor designs, Small Modular Reactors, and changing policy environments. Taken together, these factors suggest that nuclear energy remains a credible, if imperfect, contender in the pursuit of a cleaner and more secure energy future

*And the days darken round me, and the years,
Among new men, strange faces, other minds.
And slowly answer'd Arthur from the barge:
The old order changeth, yielding place to new,
And God fulfils himself in many ways,
Lest one good custom should corrupt the world.*

Alfred, Lord Tennyson, *The Passing of Arthur*

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1. Introduction & Background

We begin this paper by posing a rather loaded question: *what is the viability of nuclear energy?* To answer this question, we will first answer “build-up” questions. First, we shall lay-out how nuclear energy is generated and then answer sustainability questions – on the environmental and economic front. Finally, we will transition to answering the questions on the future of nuclear energy. But first we must establish the background of nuclear energy – from atom to the allure of clean energy.

Nuclear energy traces its roots to 1938’s Germany. Otto Hahn and Fritz Strassmann observed that uranium atoms could split into lighter elements after absorbing neutrons. Lise Meitner and Otto Frisch correctly explained the process and recognized that a very large amount of energy was released. On 2 December 1942, a team led by Enrico Fermi achieved the first controlled, self-sustaining nuclear chain reaction using Chicago Pile-1 at the University of Chicago. During World War II, nuclear research was primarily associated with the Manhattan Project and production of materials for nuclear weapons. After the war, governments increasingly investigated peaceful applications. A major milestone came on 20 December 1951, when the American Experimental Breeder Reactor I (EBR-I) in Idaho produced electricity from nuclear energy. Initially, it powered four light bulbs and later generated enough electricity for its own facility. This is generally regarded as the first generation of usable electricity from nuclear fission. The Soviet Union connected the Obninsk Nuclear Power Plant to the electricity grid in 1954. It produced around 5 MW of electricity and is generally regarded as the world’s first nuclear power station to supply electricity to a grid. Britain followed with Calder Hall, opened in 1956. Calder Hall was considerably larger and is often described as the world’s first commercial-scale nuclear power station, although it

also produced plutonium for Britain's nuclear programme. In the United States, the Shippingport Atomic Power Station in Pennsylvania began operation in 1957. It was an important early demonstration of a nuclear plant built specifically for civilian electricity generation. These plants demonstrated that reactors could provide electricity continuously rather than merely serving experimental purposes. The decade between 1960 and 1970 saw major expansion. Countries like Japan, France, Canada, Sweden, and Germany joined the USSR, the US, and the UK in the club of countries which produced a portion of their electricity through nuclear means. The 1973 oil crisis further increased interest in nuclear energy. Countries that depended heavily on imported fossil fuels saw nuclear power to improve energy security.

By the late 1970s, nuclear expansion had begun slowing in several countries.

Nuclear stations were proving increasingly expensive and complicated to construct. Regulations became stricter, construction schedules became longer, and public concern about radiation and nuclear waste increased. Incidents at Three-Mile Island (1979) and Chernobyl (1986) furthered fear in people and Governments. Nuclear plants continued generating enormous amounts of electricity during the 1990s, but relatively few new reactors were ordered in Europe and North America. China, South Korea and India increasingly viewed nuclear energy as part of their long-term electricity strategies. Meanwhile, reactor performance improved considerably. Existing reactors achieved higher capacity factors, better reliability and longer operating lifetimes. During the early 2000s, interest in nuclear power began increasing again. Concerns about climate change, fossil-fuel prices and energy security encouraged governments to reconsider nuclear energy. Then, in 2011, a massive Tsunami and earthquake caused electrical malfunctions at Fukushima in Japan. Loss of cooling caused fuel meltdowns in three reactors, while hydrogen explosions severely damaged several reactor buildings. Germany accelerated its nuclear phase-out. Japan temporarily shut down virtually its entire reactor fleet. Other countries conducted extensive safety reviews.

Today, in the decade after 2020, nuclear energy has witnessed what we may term as “cautious optimism”. Climate change, energy security, and electricity demands have driven attention to nuclear power. Some countries are extending the lives of existing reactors – while others are constructing large Generation III+ reactors. There is also substantial interest in Small Modular Reactors (SMRs), which are intended to be manufactured more systematically and constructed in smaller units than conventional gigawatt-scale plants. Researchers are additionally developing Generation IV reactors, including molten-salt reactors, sodium-cooled fast reactors and high-temperature gas reactors. Fusion energy is another research field, although commercial nuclear power

today is based on fission, not fusion. We now move on to how power is generated through nuclear means.

2. How Nuclear Power Works (Reactor Technology)

Steam, since the dawn of man's pursuit of power, has played a pivotal role in energy generation. Major efforts to generate power are simply ingenious methods to boil water! And nuclear power is no different. The following sub-sections illustrate how nuclear reactors work, and how their types vary.

2.1 The Principle of Nuclear Reactors

At their core (pun intended), nuclear reactors work on the principle of nuclear fission. Before we breakdown how this splitting happens, let us first paint a broad picture about how nuclear power gives energy. Heat generated from nuclear fission (we show how in a bit) boils water, which produces steam. This steam drives turbo generators, which generate power. That's it. It is this simple. Is it easy? To understand this, we must understand fission.

The word "*fission*" comes from the Latin *fissio*, meaning "a splitting" or "a cleaving." In the heart of the nuclear reactor (called the "core"), Uranium-235 and Plutonium-239 nuclei in the nuclear fuel are bombarded with neutrons – splitting them. This fission process produces fission fragments, additional neutrons, and a large amount of energy, which is ultimately released as heat. The newly released neutrons can then induce fission in other fuel nuclei, creating a chain reaction. These new neutrons bombard more U, and Pu, thus releasing more heat and more

neutrons. This process is known as a *chain reaction*. This is how heat is generated from fission.

The neutrons released are known as "fast neutrons", because, well, they are fast! How fast? Six to seven percent the speed of light! That's roughly 20 million meters per second. To put that in perspective, a fast neutron can circle the Earth once every 2 seconds! This speed sounds cool, but is a major issue, because at such speeds, fast neutrons generally have a much lower probability of inducing fission in U-235 than thermal neutrons. To counter this, they are slowed down by a moderator. In LWRs, such as BWRs and PWRs, ordinary light water (H₂O) serves as the moderator. PHWRs (Pressurized heavy water reactors) use Heavy water (D₂O) as moderators. Unlike LWRs, they have separate coolants and moderator circuits, coolants can be light or heavy water. We shall go over the types of reactors in detail in the next section. Now that we have shown how nuclear

reactors harness fission, a natural question arises: How is this controlled?

Chain reactions in the core of the reactor are controlled using control rods. These rods absorb neutrons. To start the reaction, these rods are removed from the core. To slow the reaction, they are gradually lowered. These rods, thus, control the reaction.

The moderator affects neutron energy, while control rods regulate the number of neutrons available to sustain the chain reaction. Interestingly, not all reactors use moderators. The next subsection details the various types of nuclear reactors.

2.2 Types of Nuclear Reactors

1. Boiling Water Reactors (BWRs)

A BWR produces steam directly by boiling the water coolant. The steam is then separated from the remaining water in “steam separators” positioned above the core, and is passed to the turbines, then condensed and recycled.

2. Pressurized Water Reactors (PWRs)

PWRs generate steam indirectly. Heat is transferred from the primary reactor coolant, which is kept liquid at high pressure, into a secondary circuit where steam is produced for the turbine.

3. Pressurized Heavy Water Reactors (PHWRs)

PHWRs use heavy water (D_2O) as moderators, in contrast to LWRs (Light Water Reactors – a combined term for BWRs, and PWRs). These may use light, or heavy water as coolants.

4. Gas-Cooled Reactors (GCRs) & Advanced Gas-cooled Reactors (AGRs)

These use CO_2 as the coolant, and graphite as the moderator. Graphite as moderator allows natural uranium (in GCRs) or very low-enriched uranium (in AGRs) fuel to be used.

5. Light Water Graphite Reactors (LWGRs)

These have enriched fuel in pressure tubes with light water as coolant. These are surrounded by graphite as a moderator. These are more commonly referred to as RBMK reactors, popularly used in the Soviet Union. The

reactor involved in the 1986 Chernobyl accident, Unit 4, was an RBMK reactor.

6. Fast Breeder Reactors (FBRs)

In these reactors, the fuel is a mix of oxides of Plutonium and Uranium, while no moderators are used. The core is usually surrounded by a ‘fertile blanket’ of uranium-238. Neutrons escaping the core are absorbed by the blanket, producing further plutonium, which is separated out during subsequent reprocessing for use as fuel. FBRs normally use liquid metal, such as sodium, as a coolant at low pressure. It is important to note that not all FBRs are the same. The description above represents a typical FBR configuration; designs vary.

7. High Temperature Gas-Cooled Reactors (HTGRs)

The HTR-PM at Shidaowan, Shandong, China, is currently the only commercial-scale HTGR power plant in operation. It entered commercial operation in December 2023. These offer an alternative to conventional designs. They use graphite as the moderator and helium as the coolant. HTGRs have ceramic-coated fuel capable of handling temperatures exceeding 1600°C and gain their efficiency by operating at temperatures of 700-950°C. The helium can drive a gas turbine directly or be used to make steam.

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2.3 A Brief Comparison of some of these Nuclear Reactors

Reactor Type	Thermal Efficiency (%)	Coolant	Moderator	Key Advantages	Key Disadvantages
Pressurized Water Reactor (PWR)	33–35%	Water	Water	High safety, used globally	High-pressure operation
Boiling Water Reactor (BWR)	32–34%	Water	Water	Simpler design, lower cost	Radiation exposure in turbines
Fast Breeder Reactor (FBR)	40–45%	Liquid Sodium	None	Efficient fuel use, breeds new fuel	Complex operation, sodium fire risk
High-Temperature Gas-Cooled Reactor (HTGR)	45–50%	Helium	Graphite	High efficiency, passive safety	Requires advanced fuel
Small Modular Reactor (SMR)	35–40%	Varies	Varies	Modular, factory-built, enhanced safety	Still under development

Source: Comparative Analysis of Nuclear Power Reactors - AHMED DASHTY

3. Sustainability & Environmental Aspects

Now that we have established how energy is generated through nuclear means – the next natural question to answer is this: *is this process sustainable?* Or better yet: *what are the broad environmental impacts of this process.* These questions are rather straightforward – but are hardly easy to answer. Nuclear energy presents a very peculiar case. Its pros are genuinely impressive, and its cons are genuinely scary.

Let us advance our discussion by discussing the pros first. The carbon emission from a nuclear plant (whilst generating energy) is very little. The IPCC has reported life-cycle nuclear emissions roughly in the range of 8–45 g CO₂-equivalent/kWh, placing nuclear among low-carbon electricity source. Most emissions occur elsewhere in the life cycle, such as uranium mining, enrichment, plant construction, fuel processing, and decommissioning.

In contrast to solar, wind, tidal, *etc.* energy generation technologies – nuclear energy is not at the behest of environmental conditions, i.e., it does not depend on the weather, making it a reliable energy generating method. Nuclear energy inherently reduces dependence on fossil fuel. Another (interesting, may we add) pro of nuclear plants is that they generate a relatively large amount of energy, in respect to the land they occupy, i.e., they are “geographically efficient”.

Finally, these plants don’t pollute the air a lot. The puffs of smoke from the cooling towers (like on the cover of this paper) – are not as grim as they appear. What you see is mainly tiny liquid water droplets formed when warm, moist air from the cooling tower condenses in cooler atmospheric air.

Naturally, not everything is *golden* in this world. Nuclear energy has its fair share of cons too. The “scariest” con is that nuclear fuel and its by-products are very dangerous. The process itself – if not undertaken with care – can be catastrophic. Chernobyl’s famous reactor 4 is a glaring example. Further, these plants depend on water a lot for various jobs like cooling. The handling of nuclear waste is a tedious process too. Given the long lifetime of nuclear fuel – their effective disposal becomes difficult. Nuclear fuel’s raw materials, such as uranium are finite, unlike the Sun, and the wind.

Clearly, nuclear energy is as beneficial – as it is tedious. The efficient and safe use of resources, coupled with active research, holds the potential to further the benefits of nuclear energy.

The next section shifts the conversation to the economic viability of nuclear energy. Afterall, everything has a cost.

4. The Economics of Nuclear Power

Amidst the flurry of debates and conversations over the wide-scale adoption of nuclear energy, a natural question emerges: “how viable is this method of energy generation in terms of its economics?” or, to put it simply: “can we afford it?”. To answer this question, two avenues emerged: (a) Cost Metrics, and (b) Systemic Challenges. Drawing comparisons between energy generation techniques is a difficult job. Why? These technologies are heterogeneous in nature. The “hard part” (if you will) lies in striking a balance between carbon emissions, waste management, system costs, decommissioning efforts, and supply chain processes. Another *crucial* factor in determining the cost of energy generation is the timing of electrical generation, which is broadly influenced by market values and system flexibility.

To draw comparisons between energy-generating technologies, we must first establish some models:

4.1 Levelized Cost of Electricity (LCOE) Model

The LCOE provides a standardized measure of the average cost of electricity generation over a plant's lifetime, making it useful for comparing technologies under a common set of assumptions. This has emerged as a dominant comparing model for various energy generation techniques. LCOE

determines the discounted lifetime average cost of generating one unit of electricity. This calculation includes capital and operating costs, as well as the average operational full-load hours of a given technology. This approach allows for a direct comparison between technologies.

The elementary formula for LCOE is:

$$LCOE = \frac{\text{Total Discounted Lifetime Costs}}{\text{Total Discounted Lifetime Energy Generated}}$$

Though the LCOE model is a good way to compare energy-generating technologies – it has its fair share of caveats. Perhaps the greatest limitation of this model is that it does not adequately capture the value of

flexibility or the broader system-level costs associated with factors such as grid integration and waste management. The LCOE provides the cost of electricity generation – but fails to capture the inherent *value* of the generated energy. To model the *value* of generated electricity, we use the LVOE model, as detailed in the following [sub]section.

4.2 Levelized Value of Electricity (LVOE) Model

Consider the following example: An energy plant produces energy *worth*

₹ 3/ kWh at 1900 hours on a cool wintry evening. The same plant, though, produces the same amount of energy, but at ₹ 20/kWh at 1400 hrs on a painful summer afternoon. Let us mediate on this for a bit. What remained constant was the quantity of electricity generated and, broadly, its generation cost. What changed was its market value. What changed? The price – and consequently, the economic value of electricity. If one were to employ the LCOE model (as described in the previous [sub] section), they would fail to capture the change in price. What really is “change in price”? Here, it implies a change in value. Electricity, during the summers, is worth more than electricity during the winters. This variation in value is captured by the

LVOE model. As the name suggests, the LVOE model focuses on the *value* of the energy generated. (*Here, market price serves as a proxy for the short-run economic value of electricity*).

An elementary formula of the LVOE model:

$$LVOE = \frac{\text{Lifetime value of energy generated}}{\text{Lifetime electricity generated}}$$

**These are discounted values*

The key point here is this: *The same amount of electricity can be produced with different economic values.*

Here is a table which consists of more such models:

Type	Metric	Conceptual expression [typical units]
Existing, Non-comprehensive	Levelized Cost of Electricity (LCOE)	Cost/Energy [MU/MWh]
	Levelized Value of Electricity (LVOE)	Value/Energy [MU/MWh]
Existing, comprehensive	Net Value of Electricity (NVOE)	(Value – Cost) / Energy [MU/MWh]
	Net Value of Capacity (NVOC)	(Value – Cost) / Capacity [MU/kW-yr]
	System LCOE	(Cost – Value) / Energy + Benchmark Price [MU/MWh]
New, comprehensive	System profitability	f(Value/Cost) [unitless]

Source Mai et al. (2021, viii)

While these metrics provide progressively more comprehensive representations of electricity economics, this review focuses primarily on LCOE and LVOE as the foundational cost and value metrics.

Now that we have established the two contemporary cost and value metrics, it is only sensible to describe how these work together to describe how feasible energy generation is. Here, we introduce the following “valuation framework”:

$LVOE > LCOE \Rightarrow \text{Economically Attractive}$

$LVOE < LCOE \Rightarrow \text{Economically Unattractive}$

4.3 Nucleo-nomics

Nuclear energy, over the past few decades, has gone through a major overhaul, and therefore, it comes as no surprise that the economics of nuclear energy has changed drastically. Perhaps the greatest change has been with the cost of nuclear energy. Over the years, as designs standardized, construction times shortened, and more efficient generation technologies were implemented, the costs associated with nuclear energy have seen a sharp decline.

These factors, along with wide adoption, increased vigilance, and little bureaucratic red tape – leading to a sense of reduced investor risk – have also led financing costs to decline. Further, a favourable inverse relation between the decrease in operating costs and increase in installed capacity – with the advent of more streamlined technologies – have led to nuclear energy becoming more affordable and “economically worth it”. In the USA, operating costs per KWh shrank by 44% between 1990 and 2003.

Another space where nuclear power has taken huge strides in is waste management, and decommissioning costs. These costs account for a fraction of operating costs, and with newer technologies, handling nuclear waste, and decommissioning reactors have

Now that we have established a few cost- and value determining models, we may finally turn our attention exclusively to the economics of nuclear power – or, as our (the authors’) combined creativity would have it, *nucleo-nomics*. (*Comedy clearly isn’t our strong suit. Why else would we be writing a paper on nuclear energy?*)

become even more economical, thus further reducing their share in the operating costs. As nuclear fuel becomes more efficient, these costs will drop further.

Below are tables from a report from an IEA report titled: Projected Costs of Generating Electricity (2020). These use the LCOE model (as discussed in section 4.1):

Technology	Number of plants / countries	Net capacity (MWe)				Overnight costs (USD/kWe)			
		Min	Mean	Median	Max	Min	Mean	Median	Max
ACAES	1 / 1	250	250	250	250	1 695	1 695	1 695	1 695
Lithium-ion battery	4 / 4	1.0	7.5	2.0	19	452	932	655	1 967
Biomass	4 / 3	0.42	18	25	30	833	2 501	1 095	6 545
Biomass (CHP)	5 / 2	0.42	130	130	358	2 538	4 689	4 689	6 545
Coal	11 / 6	138	693	722	954	800	1 897	1 785	4 382
Coal (CCUS)	4 / 2	499	641	641	650	4 490	4 572	4 572	5 991
Coal (CHP)	1 / 1	700	700	700	700	2 240	2 240	2 240	2 240
Fuel cell	4 / 2	0.003	0.20	0.20	15	2 361	4 715	4 715	6 492
Gas (CCGT)	16 / 11	471	762	743	1 372	254	823	955	1 109
Gas (CCGT, CCUS)	2 / 2	437	541	541	646	2 412	2 619	2 619	2 826
Gas (CCGT, CHP)	2 / 2	5.80	253	253	500	1 024	1 092	1 092	1 160
Gas (OCGT/int. comb.)	8 / 5	100	464	500	980	325	630	668	1 141
Gas (OCGT/int. comb., CHP)	3 / 3	35.90	119	125	195	330	707	684	1 107
Geothermal	6 / 2	5	19	19	40	3 851	6 647	6 647	10 959
Hydro (reservoir, >= 5 MW)	4 / 4	11.95	58	23	175	1 899	3 319	2 778	5 819
Hydro (reservoir, < 5 MW)	1 / 1	0.32	0.32	0.32	0.32	3 966	3 966	3 966	3 966
Hydro (run of river, >= 5 MW)	7 / 4	5	85	44	248	2 326	3 557	3 026	6 681
Hydro (run of river, < 5 MW)	18 / 4	0.02	2.54	3.00	4.80	956	3 507	2 801	7 484
Lignite	2 / 2	709	805	805	900	2 189	2 973	2 973	3 756
Lignite (CCUS)	1 / 1	570	570	570	570	6 891	6 891	6 891	6 891
Lignite (CHP)	1 / 1	2 900	2 900	2 900	2 900	1 015	1 015	1 015	1 015
Nuclear	8 / 8	950	1 234	1 137	1 650	2 157	3 606	3 370	6 920

Technology	Number of plants / countries	Net capacity (MWe)				Overnight costs (USD/kWe)			
		Min	Mean	Median	Max	Min	Mean	Median	Max
Nuclear (LTO)	4 / 4	1 000	1 000	1 000	1 000	391	504	497	629
Pumped storage	3 / 3	175	458	200	1 000	563	1 962	897	4 426
Solar PV (floating)	1 / 1	8	8	8	8	860	860	860	860
Solar PV (utility scale)	21 / 14	0.83	26	20	100	534	995	923	2 006
Solar PV (commercial)	15 / 8	0.05	0.25	0.24	0.50	846	1 065	1 085	1 357
Solar PV (residential)	15 / 8	0.004	0.01	0.01	0.02	719	1 583	1 653	2 597
Solar thermal (CSP)	4 / 2	100	125	125	150	5 238	5 857	5 857	6 475
Offshore wind	23 / 8	11.25	186	100	600	1 721	2 876	2 740	4 039
Onshore wind (>= 1 MW)	33 / 18	1	58	40	280	877	1 391	1 439	3 022
Onshore wind (< 1 MW)	11 / 1	0.01	0.19	0.19	0.90	1 782	2 852	2 852	5 539

Note: CHP = combined heat and power. CCUS = carbon capture, utilisation and storage. CCGT = Combined cycle gas turbine. OCGT = Open cycle gas turbine. LTO = long-term operation. CSP = concentrated solar power. ACAES = Adiabatic compressed air energy storage.

5. Outlook of Nuclear Energy

The previous section dealt with *nucleonomics* (the economics of nuclear energy) – establishing how nuclear energy is an economically viable source of sustainable energy – but left a rather important question open: *does a future exist where nuclear energy is the dominant energy medium?* or, to put it simply: *what is the outlook of nuclear energy?* We shall explore this very question in this section. John Steinbeck, in his book *Travels with Charley: In Search for America*, wrote: “*To find where you are going, you must know where you are*”, this simple yet profound line helps us advance our conversation about the future of nuclear energy by first asking us to pause and meditate on the state of nuclear energy today. According to the IAEA, there are 417 reactors online – having a net capacity of 397.7 GW – which produced 2,702 TWh of energy – accounting for roughly 9 per-cent of global energy generation. Outputs are at their highest levels. Nuclear energy is [clearly] flourishing. The major growth engines of this boom can be traced to Asia. China has developed standardized designs, allowing her to “mass produce” nuclear reactors (*how communist of them!*). In 2025, 9 out of 11 reactors under construction were being developed in China. Western countries on the other hand have taken a more pragmatic approach – focusing their efforts on extending the lives of existing nuclear reactors, while also restarting old reactors and designing new ones (SMRs for instance). SMRs, for lack of a better phrase, are the talk of the town today. Small Modular Reactors – as the name suggests – are small, and modular. This makes them attractive. Russia and China are at the forefront of developing SMRs – and Western countries are following suit. But conventional reactors are still the backbone of nuclear electricity generation today. But that is today. What about 2030? 2050? The IAEA projects total energy production to increase by about 40% (from 2024 production levels) by 2030 – and to 3.5 times the 2024 levels by 2050. In the bullish case, nuclear electricity production is projected to increase eightfold by 2050. The share of nuclear in total electricity production is projected to increase by 4 percentage points – on the contrary to the low case, where nuclear electricity production is projected to approximately quadruple by 2050. The share of nuclear in total electricity production is projected to increase by about one percentage point. Apart from these trends, another promising set of events arises from the political spheres. The most notable news regarding the future of nuclear energy comes from the Indian parliament which cleared the SHANTI (Sustainable Harnessing and Advancement of Nuclear Energy for Transforming India) bill, which opened limited portions of the state controlled nuclear energy sector to private players. A vast country green-lighting positive nuclear regulations is certainly a great sign for nuclear energy in the times ahead.

6. Conclusion: The Promise of *Tomorrow*?

We began this paper with a simple question: *what is the viability of nuclear energy?* – and it is only just if we end with closing this question. This section intends to do just that. Rather than being a simple “information-dump”, this section will act like an opinion piece on nuclear energy – an opinion formed after reading roughly 30 papers, and countless news articles; and of course, after authoring this work.

We are of the opinion that nuclear energy *is* viable as an energy generating technique. Zoom out enough and the entire process looks like a magic show. The splitting of atoms (building blocks we can’t even see) leading to such enormous amounts energy (while releasing miniscule carbon pollutants) – which if harnessed properly – can power the generations to come. How can anyone not be excited about this technology?

The rapid developments in reactor designs, along with legislative openness and public perception instil confidence in those (like us) who believe that nuclear energy is truly the promise of tomorrow. The research culture with respect to nuclear energy is booming. From fission tests to SMR deployments – the world truly appears to be changing. But of course, hurdles pertaining to waste disposable and overall hazards of nuclear energy still loom. What makes nuclear energy accidents most devastating is that they leave behind a set of long-lasting impacts.

From Chernobyl to Fukushima – nuclear accidents are always scary to bear, and even scarier to clean. Afterall, how does one douse invisible fires that are shooting invisible bullets, striking at the very fibre of DNA? With the advent of robust safety mechanisms – events like Chernobyl and the Three-Mile Island incident appear like warnings rather than impending reality. However, we must never join the ranks of complacency. Even water – the sustainer of life – can take it. In the end, beyond all the “*sciency*” rhetorics – the point of nuclear energy, as with any other technology – is to improve human life. And thus, safety must be at the forefront of any such conversation.

In the end, this paper – and the overall conversation regarding nuclear energy – orbits around a dream. A dream of a cleaner, secure, more sustainable world. We think that this dream is achievable. All that it demands is constant innovation, and a regard for humanity. And finally, we leave you with the words of 1947’s Man of The Year (according to TIME magazine); J. Robert Oppenheimer, who in this line captured the plight of innovation quite accurately: “*There are no secrets about the world of nature. There are secrets about the thoughts and intentions of men.*”

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We extend our sincerest gratitude to the authors of the following works. This paper would not have been possible without the giants on whose shoulders we stand.

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